Software improvements to speedyBBT, an R package for comparative judgment

Christine Stawitz, Rowland Seymour, and Heather Turner

```
Loading required package: remotes
```

```
Using GitHub PAT from the git credential store.
```

```
Skipping install of 'speedyBBT' from a github remote, the SHA1 (004b99de) has not changed si
  Use `force = TRUE` to force installation
```

```
Loading required package: knitr
```

Introduction

- Web application allows judges to input comparative judgment data
- R package for data analysis and parameter estimation
- Making the package more user-friendly and adherent with R package practices

speedyBBT

- Several forms of Bayesian Bradley-Terry model
- `speedyBBTm` uses Poly&a-Gamma latent variables to improve speed
- `BBTm.ties` allows for ties
- `BBTm` allows for comparison-specific and player-specific advantages

Requested improvements

- Use R objects to make model output use standard functions
- Improve test coverage
- Improve documentation
- Make `pkgdown` site

Data objects

```r
library(speedyBBT)
kable(forcedMarriage$comparisons[1,10])
```

Adjacency matrix

```r
kable(forcedMarriage$adjacencyMatrix[1:10,] )
```

0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
1	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Model output

```r
expA <- expm::expm(forcedMarriage$adjacencyMatrix)
prior.var <- diag(diag(expA)^-0.5) %*% expA %*% diag(diag(expA)^-0.5)
set.seed(432)
forcedMarriageModel <- speedyBBTm(
  outcome = rep(1, length(forcedMarriage$comparisons$win)),
  player1 = forcedMarriage$comparisons$win,
  player2 = forcedMarriage$comparisons$lost,
  player.prior.var = prior.var,
  n.iter = 1000,
  burn.in = 100
)
```

Model type now uses R S3 class system

- allows `speedyBBT` users to
  - use standard functions like `summary()` and `plot()`
  - use new parameter helper functions
  - integrate with `{bayesplot}`

```r
class(forcedMarriageModel)
```

New parameter helper functions

- `parameter()` returns a vector of posterior samples for a given parameter

```r
lambda <- parameter(forcedMarriageModel, "lambda", c(10,20))
hist(lambda, main = "Posterior distribution of lambda", xlab = "lambda")
```

Integration with `{bayesplot}`

- allows for easy plotting of posterior distributions and diagnostics

```r
library(bayesplot)
bayesplot::mcmc_dens(forcedMarriageModel, pars = c("lambda[10]", "lambda[20]", "lambda[30]",
```

`pkgdown` site

speedyBBT site

New vignettes

Getting Started vignette

Thank you!